AC 内二层组网直接转发无线漫游实验指导书

一、实验目的

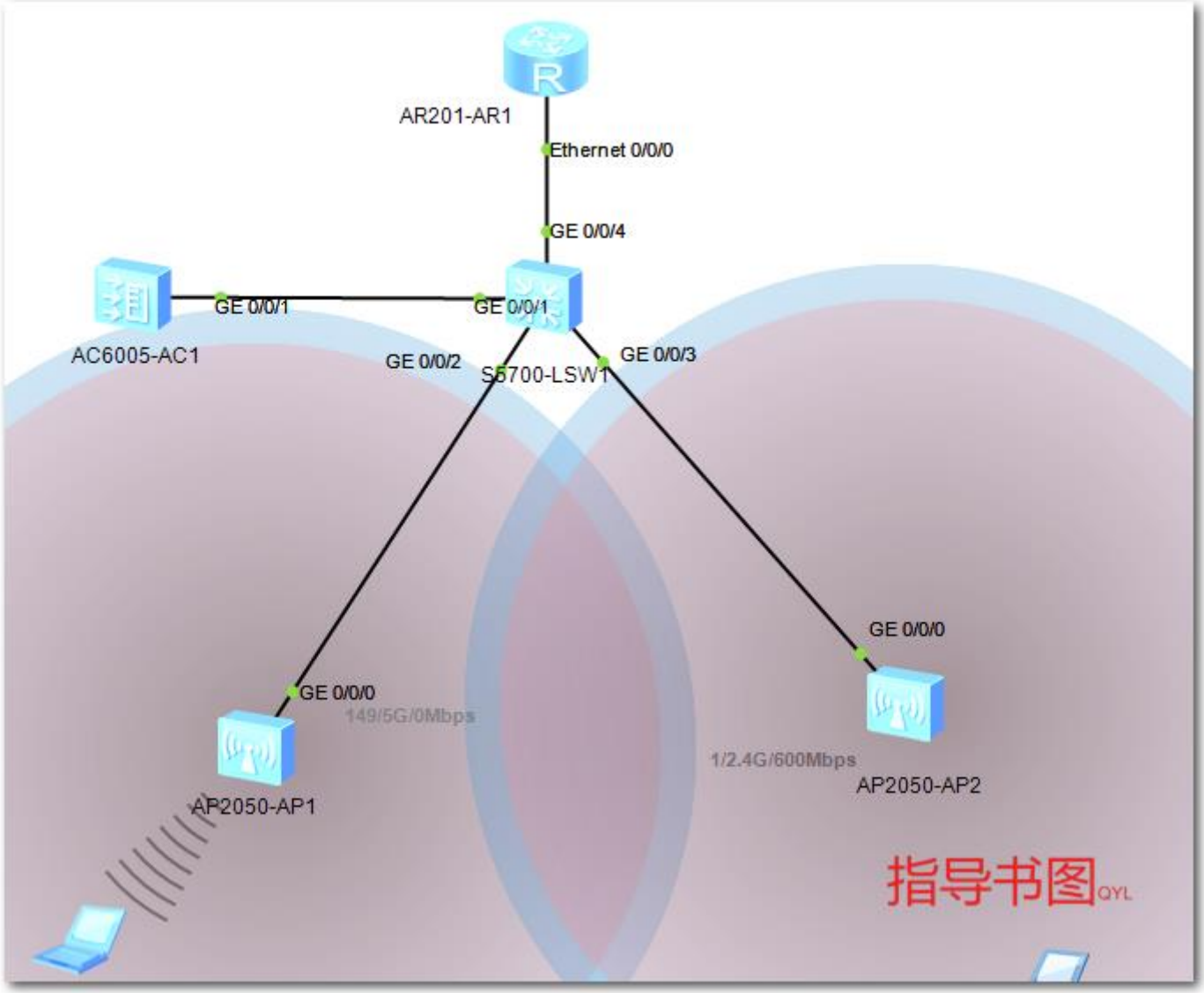
了解 AC 内二层组网直接转发无线漫游基本应用、相互关系及配置。

二、模拟器版本

ensp1.3.00.100 版本

三、实验拓扑

3.1 网络拓扑



3.2 参数规划表

配置项	规划数据
业务 VLAN	VLAN 101
管理 VLAN	VLAN 100
DHCP Server	AC 作为 DHCP 服务器为 AP 和 STA 分配 IP 地址
AP 的 IP 地址池	地址池 192.168.100.2~192.168.100.254/24
STA 的 IP 地址池	地址池 192.168.101.3~192.168.101.254/24

	网关 192.168.101.2
AC 的源接口 IP 地址	源接口地址 VLANIF100：192.168.100.1/24
AP	组名称：ap1
域管理模板	名称：domain1 国家码：CN
SSID 模板	名称：wlan-net SSID 名称：wlan-net
安全模板	名称：wlan-net 安全策略：WEP Open
VAP 模板	名称：wlan-net 转发模式：直接转发 业务 VLAN：VLAN101 引用模板：SSID 模板 wlan-net、安全模板 wlan-net

四、实验步骤

4.1 根据网络拓扑在模拟器中搭建网络；

4.2 配置路由器

```
<Huawei> system-view
[Huawei] sysname Router
[Router] vlan batch 101
[Router] interface Ethernet0/0/0
[Router-Ethernet0/0/0] port link-type trunk
[Router-Ethernet0/0/0] port trunk allow-pass vlan 101
[Router-Ethernet0/0/0] quit
[Router] interface vlanif 101
[Router-Vlanif101] ip address 192.168.101.2 24
[Router-Vlanif101] quit
```

4.3 配置 LSW1；

```
<Huawei> sysname Switch_A
[Switch_A] vlan batch 100 101
[Switch_A] interface gigabitethernet 0/0/2                    //注意：端口名要与拓扑图中的对应
[Switch_A-GigabitEthernet0/0/2] port link-type trunk
[Switch_A-GigabitEthernet0/0/2] port trunk pvid vlan 100
[Switch_A-GigabitEthernet0/0/2] port trunk allow-pass vlan 100 101
[Switch_A-GigabitEthernet0/0/2] quit
[Switch_A] interface gigabitethernet 0/0/3                    //注意：端口名要与拓扑图中的对应
[Switch_A-GigabitEthernet0/0/3] port link-type trunk
[Switch_A-GigabitEthernet0/0/3] port trunk pvid vlan 100
[Switch_A-GigabitEthernet0/0/3] port trunk allow-pass vlan 100 101
[Switch_A-GigabitEthernet0/0/2] quit
[Switch_A] interface gigabitethernet 0/0/1                    //注意：端口名要与拓扑图中的对应
[Switch_A-GigabitEthernet0/0/3] port link-type trunk
[Switch_A-GigabitEthernet0/0/3] port trunk allow-pass vlan 100 101
[Switch_A-GigabitEthernet0/0/2] quit
[Switch_A-GigabitEthernet0/0/2] quit
[Switch_A] interface gigabitethernet 0/0/4                    //注意：端口名要与拓扑图中的对应
[Switch_A-GigabitEthernet0/0/3] port link-type trunk
[Switch_A-GigabitEthernet0/0/3] port trunk allow-pass vlan 101
[Switch_A-GigabitEthernet0/0/2] quit
```

4.4 配置 **AC**：

```
<AC6605> system-view
[AC6605] sysname AC
```

[AC] vlan batch 100 101
[AC] interface gigabitethernet 0/0/1
[AC-GigabitEthernet0/0/1] port link-type trunk
[AC-GigabitEthernet0/0/1] port trunk allow-pass vlan 100 101
[AC-GigabitEthernet0/0/1] quit

4.4.1 # 创建 AP 组，用于将相同配置的 AP 都加入同一 AP 组中。

[AC] wlan
[AC-wlan-view] ap-group name ap1
[AC-wlan-ap-group-ap1] quit

4.4.2 # 创建域管理模板，在域管理模板下配置 AC 的国家码并在 AP 组下引用域管理模板。

[AC-wlan-view] regulatory-domain-profile name default
[AC-wlan-regulate-domain-default] country-code cn
[AC-wlan-regulate-domain-default] quit
[AC-wlan-view] ap-group name ap1
[AC-wlan-ap-group-ap1] regulatory-domain-profile default
Warning: Modifying the country code will clear channel, power and antenna gain configurationsof the radio and reset the AP. Continue?[Y/N]:y
[AC-wlan-ap-group-ap1] quit
[AC-wlan-view] quit

4.4.3 # 配置 AC 的源接口。

[AC] capwap source interface vlanif 100

4.4.4 配置 dhcp 服务

[AC] dhcp enable
[AC]ip pool vlan101
[AC-ip-pool-vlan101]gateway-list 192.168.101.2
[AC-ip-pool-vlan101]network 192.168.101.0 mask 255.255.255.0
[AC-ip-pool-vlan101]quit
[AC]interface Vlanif 100
[AC-Vlanif100]ip address 192.168.100.1 24
[AC-Vlanif100]dhcp select interface
[AC-Vlanif100]quit
[AC] interface vlanif 101
[AC-Vlanif101] ip address 192.168.101.1 24
[AC-Vlanif101] dhcp select global
[AC-Vlanif101] quit

4.4.5 # 在 AC 上离线导入 AP，并将 AP 加入 AP 组“ap1”中

[AC] wlan
[AC-wlan-view] ap auth-mode mac-auth
[AC-wlan-view] ap-id 0 ap-mac 00E0-FC42-5C80 //要换成你的 ap 的 mac 地址
[AC-wlan-ap-0] ap-name area_1
[AC-wlan-ap-0] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel,power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC-wlan-ap-0] quit

4.4.6 # 在 AC 上离线导入 AP2，并将 AP2 加入 AP 组“ap1”中

添加 ap2
[AC] wlan
[AC-wlan-view] ap auth-mode mac-auth
[AC-wlan-view] ap-id 1 ap-mac 00E0-FC42-5C80 //要换成 ap2 的 mac 地址
[AC-wlan-ap-0] ap-name area_2
[AC-wlan-ap-0] ap-group ap1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel,power and antenna gain configurations of the radio, Whether to continue? [Y/N]:y
[AC-wlan-ap-0] quit

4.4.7 # 将 AP 上电后，当执行命令 display ap all 查看到 AP 的“State”字段为“nor”时，表示 AP 正常上线。如下图：

```
[AC-Vlanif100]display ap all
Info: This operation may take a few seconds. Please wait for a moment.done.
Total AP information:
nor  : normal          [2]
-----------------------------------------------------------------------------
-----------
ID   MAC            Name    Group IP             Type            State STA Uptime
-----------------------------------------------------------------------------
-----------
0    00e0-fc7a-24d0 area_1 ap1   192.168.100.92 AP2050DN        nor   0   33S
1    00e0-fcfb-49f0 area_2 ap1   192.168.100.62 AP2050DN        nor   0   1M:5S
-----------------------------------------------------------------------------
-----------
Total: 2
```

指导书图

4.4.1 # 创建名为“wlan-net”的安全模板，并配置安全策略。

[AC-wlan-view] security-profile name wlan-net
[AC-wlan-sec-prof-wlan-net] security open
[AC-wlan-sec-prof-wlan-net] quit

4.4.2 # 创建名为“wlan-net”的 SSID 模板，并配置 SSID 名称为“wlan-net”。

[AC-wlan-view] ssid-profile name wlan-net
[AC-wlan-ssid-prof-wlan-net] ssid wlan-net
[AC-wlan-ssid-prof-wlan-net] quit

4.4.3 # 创建名为“wlan-net”的 VAP 模板，配置业务数据转发模式、业务 VLAN，并且引用安全模板和 SSID 模板。

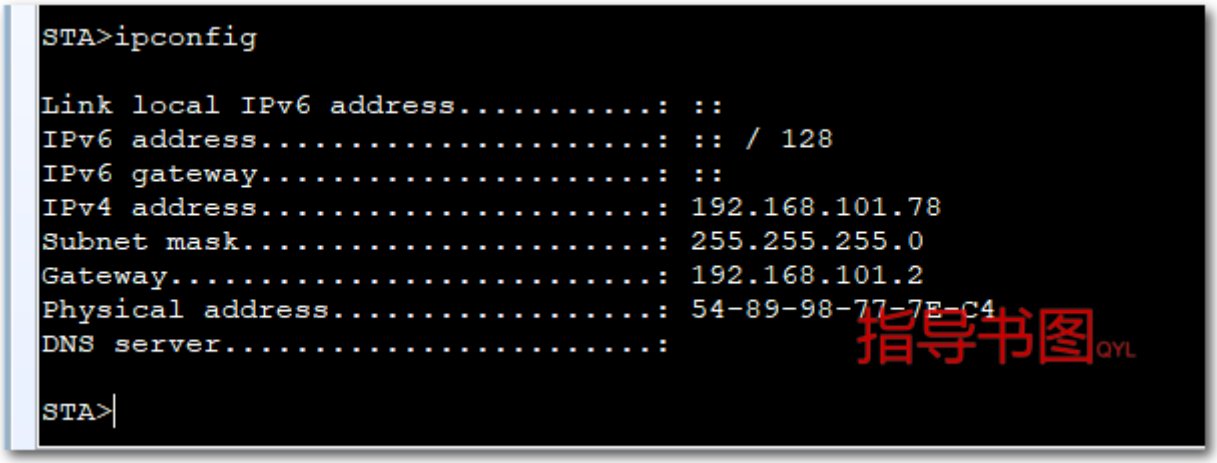
[AC-wlan-view] vap-profile name wlan-net
[AC-wlan-vap-prof-wlan-net] forward-mode direct-forward
[AC-wlan-vap-prof-wlan-net] service-vlan vlan-id 101
[AC-wlan-vap-prof-wlan-net] security-profile wlan-net
[AC-wlan-vap-prof-wlan-net] ssid-profile wlan-net
[AC-wlan-vap-prof-wlan-net] quit

4.4.4 # 配置 AP 组引用 VAP 模板，AP 上射频 0 和射频 1 都使用 VAP 模板“wlan-net”的配置。WLAN 后面数字表示 WLAN ID，表示其在 VAP 内部编号。

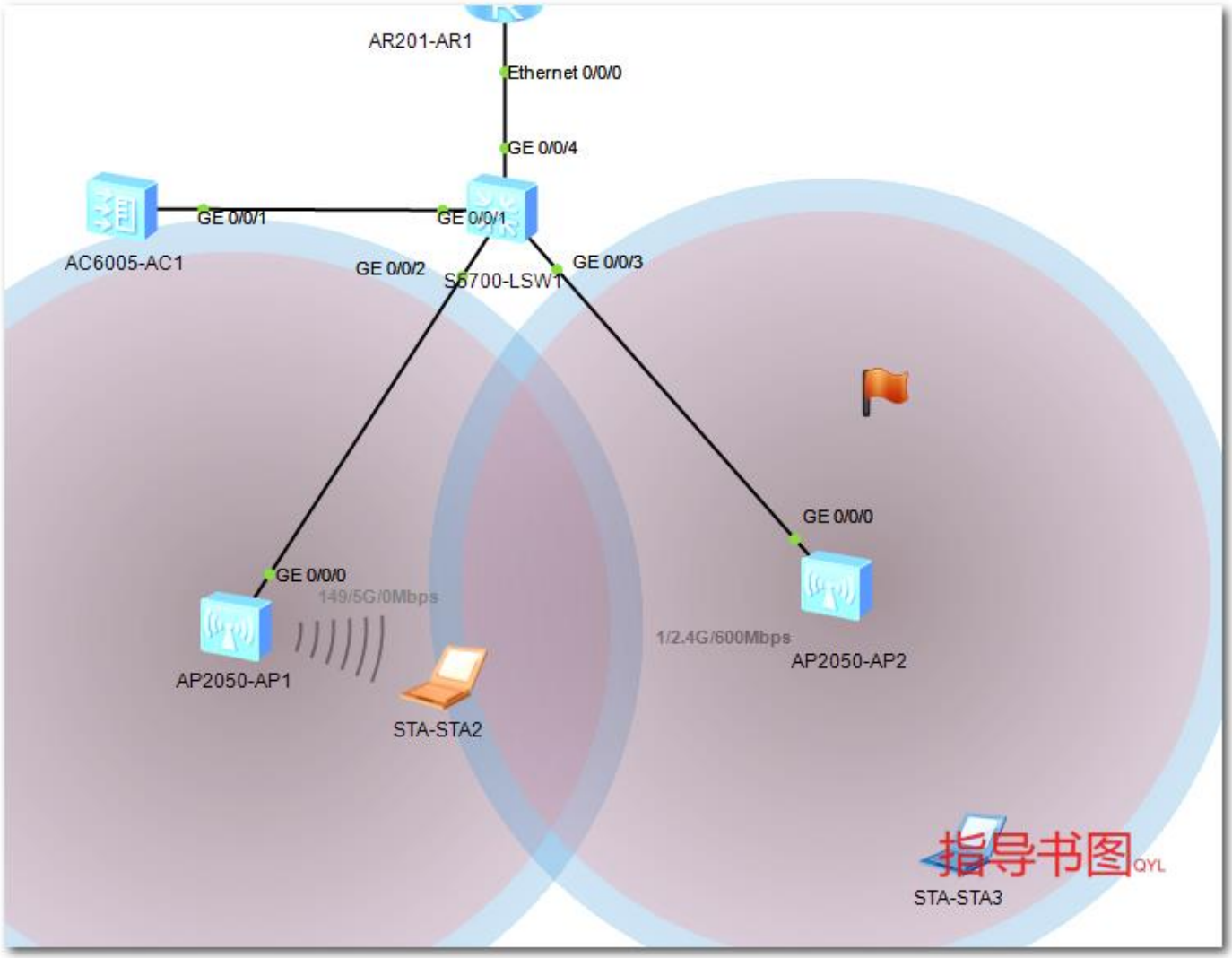
[AC-wlan-view] ap-group name ap1
[AC-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 0
[AC-wlan-ap-group-ap1] vap-profile wlan-net wlan 1 radio 1
[AC-wlan-ap-group-ap1] quit

4.5 测试网络

4.5.1 #打开 sta2，连接网络，并查看 ip，观察是否获取到 ip。



4.5.2 #右键 sta1/sta2 选择自动移动到另一个 ap 的信号范围，观察 sta1/sta2 是否能正常连接。



4.5.3 #在 AC 上执行命令 display station roam-track sta-mac 5489-98D4-5125 //红色为 sta 的 mac 地址。观察 sta 的漫游记录。

```
<AC>display station roam-track sta-mac 5489-98D4-5125
Access SSID:wlan-net
Rx/Tx: link receive rate/link transmit rate(Mbps)
z: Zero Roam c:PMK Cache Roam r:802.11r Roam
-------------------------------------------------------------------------------
L2/L3           AC IP                   AP name                 Radio ID
BSSID           TIME                    In/Out RSSI             Out Rx/Tx
-------------------------------------------------------------------------------
--              192.168.100.1           area_2                  0
00e0-fcfb-49f0  2024/09/23 15:13:05     -95/-95                 0/0
L2              192.168.100.1           area_1                  0
00e0-fc7a-24d0  2024/09/23 15:26:07     -95/-95                 0/0
L2              192.168.100.1           area_2                  0
00e0-fcfb-49f0  2024/09/23 15:26:09     -95/-95                 0/0
L2              192.168.100.1           area_1                  0
00e0-fc7a-24d0  2024/09/23 15:26:12     -95/-95                 0/0
L2              192.168.100.1           area_2                  0
00e0-fcfb-49f0  2024/09/23 15:26:14     -95/-                   -/-
-------------------------------------------------------------------------------
Number: 4
<AC>
```

指导书图

五、实验需注意问题

1.关闭 ensp 程序前，需注意保存设备配置（在设备命令行中输入 save）以及保存 topo 文件（点击保存按钮）。